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Exercise-I Classification of elements and Periodicity in Properties - Periodic Table

1. Among P, S, Cl, F most and least negative ΔH_{eg} will be respectively of:
(1) Cl, P (2) P, Cl
(3) Cl, F (4) F, Cl
CPT314

2. Which order for atomic radii is incorrect?
(1) $H^- > Li^+ \geq Mg^{+2} > Al^{+3}$
(2) $MnO_2 > KMnO_4$
(3) $O^{-2} > F^- > Na^+ > Mg^{+2}$
(4) $B > Al \simeq Ga < In \simeq Tl$
CPT315

3. Which is not incorrect for acidic strength?
(1) $H_2S < H_2Se < H_2Te$
(2) $HClO_4 > HClO_3 > HClO_2 > HClO$
(3) $P_4O_{10} > SiO_2$
(4) All are correct
CPT316

4. Select the correct matching in column-I with column-II.

Column-I (Atomic number)	**Column-II** (IUPAC official name)
(a) 101	(P) Mendelevium
(b) 105	(Q) Dubnium
(c) 107	(R) Bohrium
(d) 103	(S) Lawrencium

(1) (a)-P, (b)-Q, (c)-R, (d)-S
(2) (a)-Q, (b)-P, (c)-S, (d)-R
(3) (a)-R, (b)-P, (c)-Q, (d)-S
(4) (a)-P, (b)-Q, (c)-S, (d)-R
CPT317

5. In which of the following pairs, first member has higher first IP
(a) N, O (b) B, Be
(c) Al, Ga (d) F, Cl
(e) Zn, Ga (f) F^-, Cl^-
Correct option is
(1) a, c, f, d (2) a, d, e
(3) b, d, e, f (4) a, d, e, f
CPT318

6. The smallest cation and the smallest anion are respectively
(1) H^+ and H^- (2) H^+ and F^-
(3) Li^+ and H^- (4) Li^+ and F^-
CPT319

7. Which of the following order of IP is incorrect
(1) $Na^+ > Mg^+$ (2) $Mg^{+2} > Mg^+$
(3) $He > Li^+$ (4) $Be > B$
CPT320

8. Correct order of increasing atomic size is
(1) N < F < Si < P (2) F > N < P < Si
(3) F < N < P < Si (4) F < N < Si < P
CPT321

9. Which of the following is the most basic oxide?
(1) SeO_2 (2) Al_2O_3
(3) Sc_2O_3 (4) Bi_2O_3
CPT322

10. Electronic configurations of some elements are given in column I and their electron gain enthalpies are given in column II. Match the electronic configuration with electron gain enthalpy.

Column (I) Electronic Configuration	**Column (II)** Electron gain enthalpy/kJ mol^{-1}
(i) $1s^2\ 2s^2\ 2p^1$	(A) –53
(ii) $1s^2\ 2s^2\ 2p^6\ 3s^1$	(B) –328
(iii) $1s^2\ 2s^2\ 2p^5$	(C) –141
(iv) $1s^2\ 2s^2\ 2p^4$	(D) +48

(1) i → D, ii → A, iii → B, iv → C
(2) i → A, ii → B, iii → C, iv → D
(3) i → D, ii → B, iii → A, iv → C
(4) i → D, ii → C, iii → A, iv → B
CPT323

11. Order of atomic radius is correct of the elements given below ?
(1) Fe ≃ Co ≃ Ni (2) Ni > Co > Fe
(3) Co > Ni > Fe (4) Co > Fe > Ni
CPT324

12. The correct values of ionization enthalpies (in kJ mol^{-1}) of Si, P, Cl and S respectively are:-
(1) 786, 1012, 999, 1256
(2) 1012, 786, 999, 1256
(3) 786, 1012, 1256, 999
(4) 786, 999, 1012, 1256

CPT325

13. The correct order of second ionization potential of C, N, O and F is :-
(1) C > N > O > F (2) O > N > F > C
(3) O > F > N > C (4) F > O > N > C

CPT326

14. Element 'X' having electronic configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$ forms compound with Ca. The compound is :-
(1) Ca_2X_3 (2) Ca_3X
(3) Ca_3X_2 (4) CaX

CPT327

15. The ionic radius of Cr is minimum in which of the following compounds ?
(1) CrF_3 (2) $CrCl_3$
(3) Cr_2O_3 (4) K_2CrO_4

CPT328

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Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	4	1	2	2	3	3	4	1	1	3	3	3	4

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Exercise-II Classification of elements and Periodicity in Properties - Periodic Table

1. In which of the following options order of arrangement does not agree with the variation of property indicated against it?
(1) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$ (increasing ionic size)
(2) B < C < N < O (increasing first ionisation enthalpy)
(3) I < Br < F < Cl (increasing electron gain enthalpy)
(4) Li < Na < K < Rb (increasing metallic radius)
CPT329

2. **Statement-1 :** Acidic nature of the following Hydride of group 16 element in the order $H_2O > H_2S > H_2Se > H_2Te$.
Statement-2 : As the size of atom increases bond strength decreases.
(1) Both statement are correct and statement-2 is the correct explanation of statement-1.
(2) Both statement are correct but statement-2 is not the correct explanation of statement-1.
(3) Statement-1 is true but statement-2 is false.
(4) Statement-1 is false but statement-2 is true.
CPT330

3. **Assertion (A) :-** Mn_2O_7 is more acidic than MnO_2.
Reason (R) :- Higher the positive charge. Higher will be EN more will be acidic nature.
(1) Both (A) and (R) correct and (R) is correct explanation of (A).
(2) Both (A) and (R) correct and (R) is not correct explanation of (A).
(3) (A) is correct but (R) is incorrect.
(4) Both (A) and (R) is incorrect.
CPT331

4. In which of the following arrangements, the sequence is not strictly according to the property written against it ?
(1) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power
(2) HF < HCl < HBr < HI : increasing acid strength
(3) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing Lewis basic strength
(4) B < C < O < N : increasing first ionisation enthalpy
CPT332

5. Which one of the following arrangement represent incorrect order.
(1) O < S < F < Cl Electron gain enthalpy
(2) Al < Mg < S < P Ionisation energy
(3) F > Cl > O > S Electronegativity
(4) $Mg^{+2} < Na^{+} < F^{-} < O^{-2}$ Ionic size
CPT333

6. Which of the following order is wrong :-
(1) $NH_3 < PH_3 < AsH_3$ — Acidic
(2) Li < Be < B < C — First IP
(3) $Al_2O_3 < MgO < Na_2O < K_2O$ — Basic
(4) $Li^{+} < Na^{+} < K^{+} < Cs^{+}$ — Ionic Radius
CPT334

7. Following are configuration of 4 atom :-
P = $(Ne)3s^2 3p^3$ Q = $(Ar)3d^{10}4s^2 4p^3$
R = $(He)2s^2 2p^5$ S = $(Ne)3s^1$
Incorrect statement is :-
(1) EA : Q > R (EA = electron affinity)
(2) EN : R > P (EN = electro negativity)
(3) IP : P > S (IP = Ionisation potential)
(4) Atomic radius : S > R
CPT335

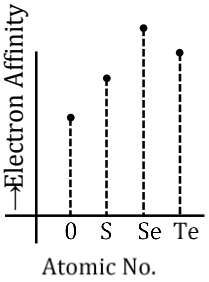
8. Which of the following statements are not correct:-
(1) The electron affinity of 'Si' is greater than that of C
(2) BeO is amphoteric while B_2O_3 is acidic
(3) The ionisation energy of 'Tl' is less than that of Al
(4) The ionisation energy of elements of 'Cu' group is less than that of the respective elements of Zn-group
CPT336

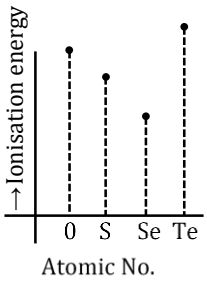
9. Among the following statement identify incorrect statement?
(1) O has least electron affinity in its group.
(2) I has largest E.A. among halogen.
(3) S^{-} has higher I.E than O^{-}
(4) None of these
CPT337

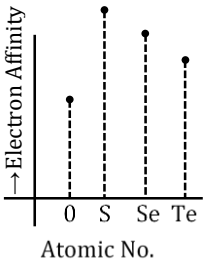
10. The compound X – O – H is likely to act as a base, if compared to hydrogen, X has :-
(1) Higher ionization potential
(2) Lower electron negativity
(3) Higher electronegativity
(4) Lower atomic radius

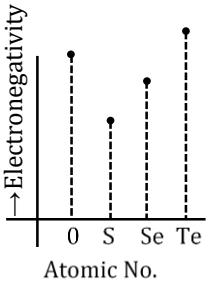
CPT338

11. Select the correct graph for given property for chalcogen family.

(1)


(2)


(3)


(4)


CPT339

12. Which of the following order is valid for Li < B < Be < C < O < N < F < Ne
(1) IE (2) EGE
(3) EN (4) size

CPT340

13. **Statement-1 :-** All inert gas have ns^2np^6 configuration.
Statement-2 :- Inert gas have fully filled outermost shell.
(1) Statement-1 is incorrect while statement-2 is correct.
(2) Both statement is incorrect.
(3) Both statement is correct.
(4) Statement-1 is correct but statement-2 is incorrect.

CPT341

14. Which is not correct order for the stated property?
(1) Ba > Sr > Mg : Atomic radius
(2) F > O > N : First ionisation energy
(3) Cl > F > I : Electron affinity
(4) O > Se > Te : Electronegativity

CPT342

Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Answer	**2**	**4**	**1**	**3**	**3**	**2**	**1**	**3**	**2**	**2**	**3**	**1**	**1**	**2**	

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Exercise – I Chemical Bonding and Molecular Structure – Chemical Bonding

1. Determine the incorrect order of bond angle?
(1) $NH_3 < NF_3 < NCl_3$
(2) $OF_2 < OH_2 < OCl_2$
(3) $SF_2 < SCl_2 < SBr_2 < SI_2$
(4) $ClO_2 > ClO_2^{-1} > OCl_2$

CCB479

2. There are some species given below :-
(a) O_2^+ (b) CO (c) B_2
(d) O_2^- (e) NO^+ (f) He_2^+
(g) C_2^{+2} (h) CN^- (i) N_2^-
Total no. of species which have their fractional bond order.
(1) 3 (2) 4
(3) 5 (4) 6

CCB480

3. The correct order of N–O bond length is
(1) $NO_3^- > NO_2^+ > NO_2^-$
(2) $NO_3^- > NO_2^- > NO_2^+$
(3) $NO_2^+ > NO_3^- > NO_2^-$
(4) $NO_2^- > NO_3^- > NO_2^+$

CCB481

4. Which of the following have same bond order:-
(I) CO (II) CN^-
(III) O_2^+ (IV) NO^+
(1) I, II, III (2) I, II, IV
(3) I, III, IV (4) II, III, IV

CCB482

5. The correct stability order of N_2 and its given ions is
(1) $N_2 > N_2^+ > N_2^- > N_2^{2-}$
(2) $N_2^- > N_2^+ > N_2 > N_2^{2-}$
(3) $N_2^+ > N_2^- > N_2 > N_2^{2-}$
(4) $N_2 > N_2^+ = N_2^- > N_2^{2-}$

CCB483

6. A compound which leaves behind no residue on heating is
(1) $Cu(NO_3)_2$ (2) KNO_3
(3) NH_4NO_3 (4) None of these

CCB484

7. Which of the following molecule is polar and non-planar
(1) XeF_4 (2) XeF_5^- (3) CH_2F_2 (4) ClF_3

CCB485

8. Dipole moment of NH_3 is more than NF_3 because
(1) N–F bond is more polar than N–H bond
(2) NH_3 is pyramidal while NF_3 is planar
(3) In NH_3 orbital dipole due to lone pair is in the same direction as the resultant dipole moment of N–H bonds while in NF_3 orbital dipole due to lone pair is in opposite direction of the resultant dipole moment of N–F bonds
(4) None of these

CCB486

9. Which of the following is polar
(1) p-dichlorobenzene
(2) trans-1-chloropropene
(3) boron tri fluoride
(4) xenon tetra fluoride

CCB487

10. Which of the following attraction is strongest?
(1) Cl^- ⋯ H—Cl
(2) $CHCl_3$ ⋯ $CHCl_3$
(3) CCl_4 ⋯ H_2O
(4) Cl^- ⋯ H_2O

CCB488

11. Statement-1 : The magnitude of H-bonding depends on the physical state of compound.
Statement-2 : Hydrogen bonding have strong influence on the structure and properties of the compound.
(1) Statement-1 is correct, Statement-2 is incorrect.
(2) Statement-1 is incorrect, Statement-2 is correct.
(3) Both statement are incorrect.
(4) Both statement is correct.

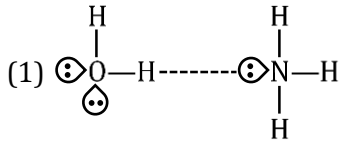
CCB489

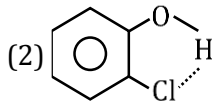
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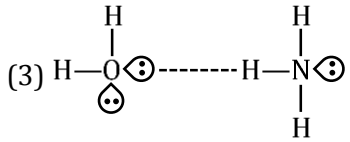
12. The correct order of dipole moment is:
(1) $CH_4 < NF_3 < NH_3 < H_2O$
(2) $NF_3 < CH_4 < NH_3 < H_2O$
(3) $NH_3 < NF_3 < CH_4 < H_2O$
(4) $H_2O < NH_3 < NF_3 < CH_4$

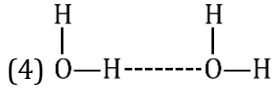
CCB490

13. Which of the following is not a best representation of the H-bond?

(1) 

(2) 

(3) 

(4) 

CCB491

14. Which of the following carbonate is maximum stable to heat?
(1) $CaCO_3$ (2) Li_2CO_3
(3) Na_2CO_3 (4) $BaCO_3$

CCB492

15. In which of the following molecule/ion all the bonds are not equal?
(1) XeF_4 (2) BF_4^-
(3) C_2H_4 (4) SiF_4

CCB493

16. The incorrect order of solubility in water :-
(1) $Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$
(2) $Li_2CO_3 < Na_2CO_3 < K_2CO_3$
(3) AgF<AgCl<AgBr
(4) $BaSO_4 < MgSO_4$

CCB494

17. The hybridisation of P in phosphate ion (PO_4^{3-}) is the same as in:
(1) I in ICl_4^- (2) S in SO_3
(3) N in NO_3^- (4) S in SO_3^{2-}

CCB495

18. Which of the following has regular tetrahedral shape.
(1) BF_4^- (2) XeF_4
(3) SF_4 (4) ClF_4^-

CCB496

19. The correct sequence of decrease in bond angles of following hydrides is:
(1) $NH_3 > PH_3 > AsH_3 > SbH_3$
(2) $SbH_3 > AsH_3 > PH_3 > SbH_3$
(3) $SbH_3 > AsH_3 > PH_3 > NH_3$
(4) $PH_3 > NH_3 > AsH_3 > SbH_3$

CCB497

20. Compare F — Î — O and F_{axial} — Î — F_{axial} bond angle in IOF_3 molecule:
(1) F—Î—O > F_{axial}—Î—F_{axial}
(2) F_{axial}—Î—F_{axial} > F—Î—O
(3) F_{axial}—Î—F_{axial} = F—Î—O
(4) None of these

CCB498

21. Which of the following order of energies of molecular orbitals of N_2 is correct?
(1) $(\pi_{2p_y}) < (\sigma_{2p_z}) < (\pi^*_{2p_x}) \approx (\pi^*_{2p_y})$
(2) $(\pi_{2p_y}) > (\sigma_{2p_z}) > (\pi^*_{2p_x}) \approx (\pi^*_{2p_y})$
(3) $(\pi_{2p_y}) < (\sigma_{2p_z}) > (\pi^*_{2p_x}) \approx (\pi^*_{2p_y})$
(4) $(\pi_{2p_y}) > (\sigma_{2p_z}) < (\pi^*_{2p_x}) \approx (\pi^*_{2p_y})$

CCB499

Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	2	2	1	3	3	3	2	4	4	1	3	3	3
Question	16	17	18	19	20	21									
Answer	3	4	1	1	2	1									

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Exercise – II Chemical Bonding and Molecular Structure – Chemical Bonding

1. Select correct order out of given options :-
(1) $BeCO_3 < BaCO_3 \rightarrow$ Covalent character
(2) $BeO > SrO \rightarrow$ Lattice energy
(3) $Be^{2+} < Li^+ \rightarrow$ Hydration energy
(4) $Be^{2+}_{(aq.)} > Li^+_{(aq.)} \rightarrow$ Ionic Mobility
CCB500

2. Correct order is :-
(1) $SOF_2 > SOCl_2 > SOBr_2$ (B.A.) : $(X-S-X)_{X=F/Cl/Br}$
(2) $SF_4 > XeF_4$ (Dipole moment)
(3) $BF_3 < BCl_3 < BBr_3$ (Bond angle)
(4) $PO_4^{3-} < SO_4^{2-} < ClO_4^-$ (Bond length) (M–O)
CCB501

3. Which statement is correct in the following :-
(1) $NH_3 > NF_3$: bond angle
(2) $NH_3 > NF_3$: reactivity towards lewis acid
(3) $NH_3 > NF_3$: Dipole moment
(4) All are correct
CCB502

4. **Statement-1 :** p-hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.
Statement-2 : o-hydroxybenzoic acid has intermolecular hydrogen bonding.
(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(3) Statement-1 is True, Statement-2 is False.
(4) Statement-1 is False, Statement-2 is False.
CCB503

5. Which is correct?
(1) PbS > ZnS (Solubility)
(2) $Li_2CO_3 > Na_2CO_3$ (Thermal stability)
(3) NaF > KF (Lattice energy)
(4) $BaSO_4 > MgSO_4$ (Solubility)
CCB504

6. Which of the following order is not correct :-
(1) $SO_4^{-2} = PO_4^{-3} = ClO_4^-$ (Bond angle)
(2) $OCl_2 < ClO_2$ (Bond angle)
(3) $ZnCl_2 < CdCl_2 < HgCl_2$ (ionic character)
(4) $CH_3–Cl > CH_3F > CH_3–Br > CH_3–I$ (Dipole moment)
CCB505

7. Which is not correct?
(1) White vitriol and epsom salt are isomorphous
(2) Thermal stability
$\rightarrow BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3$
(3) Solubility
$\rightarrow NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$
(4) Melting point $\rightarrow Al_2O_3 < MgF_2$
CCB506

8. Which is correct?
(1) Ionic mobility in aqueous medium
$\rightarrow Li^+ < Na^+ < K^+ < Rb^+$
(2) Covalant character
$\rightarrow KCl > CaCl_2 > AlCl_3 > SnCl_4$
(3) Boiling point order
$\rightarrow H_2O > H_2Se > H_2Te > H_2S$
(4) Dipole – dipole attraction $\rightarrow KCl + H_2O$
CCB507

9. Consider the following order
(1) $PH_3 > NH_3 > AsH_3$ (basic character)
(2) $PH_3 > NH_3 > AsH_3$(boiling point)
(3) $HOCl > HClO_2 > HClO_3 > HClO_4$ (oxidising property)
(4) HF > HCl > HBr > HI (acidic)
(5) $H_2O > H_2S > H_2Se$ (bond angle)
(6) $H_2SO_4 > H_3PO_4 > H_2CO_3$ (acidic character)
correct order(s) are
(1) 1,2,4,5 (2) 2,3,5,6 (3) 3,5,6 (4) 4,5,6
CCB508

10. Which of the following statement is incorrect
(1) Removal of an electron is easier from O_2 in comparison to O_2^{+2}
(2) In the double bond of C_2 molecule, both are π-bonds
(3) NO is more stable than NO^+
(4) NO_2^+ and CO_2 are isoelectronic and isostructural
CCB509

11. Match the column

Column I	**Column II**
(a) C_2H_2	(P) sp^3d hybridisation
(b) SO_2	(Q) sp^3 hybridisation
(c) I_3^-	(R) sp^2 hybridisation
(d) NH_4^+	(S) sp hybridisation

(1) (a) S (b) P (c) R (d) Q
(2) (a) P (b) S (c) R (d) Q
(3) (a) S (b) R (c) P (d) Q
(4) (a) R (b) S (c) P (d) Q
CCB510

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12. Which of the following is correct statement ?
(a) $AlCl_3$ is conducting in fused state
(b) Mobility of Li^+ ion in water is greater than Cs^+ ion
(c) MCl_2 is more volatile than MCl_4
(d) $BeSO_4$ is more soluble in water than $BaSO_4$
(1) a, b (2) b, c, d (3) b, d (4) Only d
CCB511

13. Which of the following is not true about H_2O molecule ?
(1) The molecule has $\mu = 0$
(2) The molecule can act as a base
(3) Shows abnormally high boiling point in comparison to the hydrides of other elements of oxygen group
(4) The molecule has a bent shape
CCB512

14. If AB_4^n, type species are tetrahedral, then which of the following is incorrectly matched?

	A	B	n
(1)	Xe	O	Zero
(2)	Se	F	Zero
(3)	P	O	–3
(4)	N	H	+1

CCB513

15. Which of the following statement are **correct**?
(a) In PCl_5 hybridisation is sp^3d and it has a trigonal pyramidal structure.
(b) The angle between the P-Cl bonds is 90º, which is same for all the P and Cl bond present in PCl_5
(c) The bond length of P-Cl in axial position is higher than in equitorial position
(d) PCl_5 have zero dipole moment
Choose the correct option.
(1) a & b (2) b & c (3) c & d (4) d & a
CCB514

16. Which of the following molecule/species are iso-structural with N_3^- ion?
(1) I_3^+ (2) I_3^- (3) NH_2^- (4) HCO_2^-
CCB515

17. Which of the following molecule have both $p\pi - p\pi$ and $p\pi - d\pi$ bonding?
(1) ClO_2^+ (2) NO_2^+ (3) SO_3^{2-} (4) ClO_4^-
CCB516

18. Statement-1 : dipole moment of CH_3–Cl is more than CH_3–F.
Statement-2 : CH_3–Cl have more ΔEN than CH_3–F.
(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(3) Statement-1 is True, Statement-2 is False.
(4) Statement-1 is False, Statement-2 is False.
CCB517

19. The true statement from the following is -
(1) PH_5 , NCl_5 and $BiCl_5$ do not exist
(2) I_3^- has bent geometry
(3) XeF_4 is polar molecule
(4) O_2 and O_2^{-2} has same bond order
CCB518

20. Which order are correct?
(I) Thermal stability :
$BeSO_4 < MgSO_4 < CaSO_4 < BaSO_4$
(II) Basic nature :
$ZnO > BeO > MgO > CaO$
(III) Solubility in water :
$LiOH > NaOH > KOH > RbOH > CsOH$
(IV) Melting point :
$NaCl > KCl > RbCl > CsCl > LiCl$
(1) (I), (IV) (2) (I), (II) and (IV)
(3) (II), (III) (4) All correct
CCB519

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**2**	**4**	**4**	**3**	**3**	**4**	**1**	**3**	**3**	**3**	**4**	**1**	**2**	**3**
Question	16	17	18	19	20										
Answer	**2**	**1**	**3**	**1**	**1**										

Exercise - I p-Block Elements - Group 13 to 18

1. How many bridging oxygen atoms are present in P_4O_{10}?
(1) 6 (2) 4 (3) 2 (4) 5
CPB393

2. Which of the following phosphorus is the most reactive?
(1) Scarlet phosphorus
(2) White phosphorus
(3) Red phosphorus
(4) Violet phosphorus
CPB394

3. Mixture used in Holme's signal is?
(1) CaC_2 and $CaCl_2$
(2) $CaCl_2$ and Ca_3P_2
(3) CaC_2 and Ca_3N_2
(4) CaC_2 and Ca_3P_2
CPB395

4. HNO_2 acts as
(1) acid (2) oxidising agent
(3) reducing agent (4) All
CPB396

5. Bleaching action of SO_2 is due to–
(1) reduction (2) oxidation
(3) hydrolysis (4) acidic nature
CPB397

6. Red and white phosphorus are similar in
(1) smell
(2) solubility in CS_2
(3) Hybridisation of P
(4) Stability
CPB398

7. In which of the following oxy acid of sulphur, sulphur atoms has different oxidation states
(1) $H_2S_4O_6$ (2) $H_2S_2O_8$
(3) $H_2S_2O_4$ (4) All
CPB399

8. The catalyst used in the manufacture of ammonia by Haber's process is:
(1) Pt (2) Fe_2O_3
(3) $CuCl_2$ (4) V_2O_5
CPB400

9. Industrial preparation of nitric acid by Ostwald's process involves:
(1) Oxidation of NH_3
(2) Reduction of NH_3
(3) Hydrogenation of NH_3
(4) Hydrolysis of NH_3
CPB401

10. When zinc reacts with very dilute nitric acid it produces:
(1) NH_4NO_3 (2) NO
(3) NO_2 (4) H_2
CPB402

11. Atoms in P_4 molecule of white phosphorus are arranged regularly in the following way:
(1) At the corners of a cube
(2) At the corners of an octahedron
(3) At the corners of a tetrahedron
(4) At the centre and corners of a tetrahedron
CPB403

12. The most thermodynamically stable allotropic form of phosphorus is:
(1) Red (2) White
(3) Black (4) Yellow
CPB404

13. Which of the following can be hydrolysed?
(1) TeF_6 (2) NCl_3
(3) SF_6 (4) Both 1 & 2
CPB405

14. Which of the following methyl diboranes does not exist ?
(1) $B_2H_4(CH_3)_2$ (2) $B_2H_5(CH_3)$
(3) $B_2H_2(CH_3)_4$ (4) $B_2H(CH_3)_5$
CPB406

15. Which of the following does not form during hydrolysis of XeF_6?
(1) XeO_3 (2) $XeOF_4$
(3) XeO_2F_2 (4) $XeOF_3$
CPB407

16. Beryl is a type of
(1) chain silicate (2) cyclic silicate
(3) sheet silicate (4) 3-D silicate
CPB408

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17. The chain length of silicones can be controlled by
(1) $(CH_3)_3SiCl$
(2) Addition of Cu powder
(3) Elevation of temperature
(4) None of these

CPB409

18. Glass is soluble in
(1) aqua regia (2) H_2SO_4
(3) HF (4) $HClO_4$

CPB410

19. Borax bead test is not given by
(1) An aluminium salt
(2) A cobalt salt
(3) A copper salt
(4) A nickel salt

CPB411

20. Which of the following silicate is called disilicate?
(1) Orthosilicate
(2) Pyrosilicate
(3) Single-chain silicate
(4) None of these

CPB412

21. Thallium shows different oxidation states because:
(1) of its high reactivity
(2) of inert pair of electrons
(3) of its amphoteric nature
(4) It is a transition metal

CPB413

22. H_3BO_3 is :
(1) Monobasic and weak Lewis acid
(2) Monobasic and weak bronsted acid
(3) Monobasic acid and strong Lewis acid
(4) Tribasic acid and weak bronsted acid

CPB414

23. Carborundum is:
(1) Al_2O_3 (2) SiC
(3) BF_3 (4) B_4C

CPB415

24. The stability of dihalides of Si, Ge, Sn, and Pb increase stability in the sequence:
(1) $PbX_2 < SnX_2 < GeX_2 < SiX_2$
(2) $GeX_2 < SiX_2 < SnX_2 < PbX_2$
(3) $SiX_2 < GeX_2 < PbX_2 < SnX_2$
(4) $SiX_2 < GeX_2 < SnX_2 < PbX_2$

CPB416

25. C—C bond length is maximum in:
(1) Diamond (2) Graphite
(3) Napthalene (4) Fullerene

CPB417

26. Catenation i.e., linking of similar atoms depends on size and electronic configuration of atoms. The tendency of catenation in group 14 elements follows the order
(1) C > Si > Ge > Sn
(2) C > Si > Ge ≈ Sn
(3) Si > C > Sn > Ge
(4) Ge > Sn > Si > C

CPB418

27. The possible oxidation state of Tl are:
(1) + 1 and +2 (2) +2 and +3
(3) +1 and – 1 (4) + 1 and + 3

CPB419

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**2**	**4**	**4**	**1**	**3**	**1**	**2**	**1**	**1**	**3**	**3**	**4**	**4**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26	27			
Answer	**2**	**1**	**3**	**1**	**2**	**2**	**1**	**2**	**4**	**1**	**2**	**4**			

Exercise - II p-Block Elements - Group 13 to 18

1. $P+Cl_2 \rightarrow A$,
$P + \text{Excess } Cl_2 \rightarrow B$
Hydrolysis products of A and B are respectively
(1) H_3PO_2, H_3PO_3
(2) H_3PO_4, H_3PO_3
(3) H_3PO_3, H_3PO_4
(4) H_3PO_2, H_3PO_4

CPB420

2. Which of the following are peroxy-acid of sulphur
(1) H_2SO_5 and $H_2S_2O_8$
(2) H_2SO_5 and $H_2S_2O_7$
(3) $H_2S_2O_7$ and $H_2S_2O_8$
(4) $H_2S_2O_6$ and $H_2S_2O_7$

CPB421

3. An example of tetrabasic acid is :-
(1) Orthophosphorus acid
(2) Orthophophoric acid
(3) Metaphosphoric acid
(4) Pyrophosphoric acid

CPB422

4. Which of the following compounds are the common products obtained in the hydrolysis of XeF_6 and XeF_4?
(1) XeO_2F_2
(2) HF
(3) XeO_3
(4) Both (2) and (3)

CPB423

5. Identify the incorrect statement among the following:
(1) Ozone reacts with SO_2 to give SO_3
(2) Silicon reacts with NaOH(aq.) in the presence of air to give Na_2SiO_3
(3) Cl_2 reacts with excess of NH_3 to give N_2 and NH_4Cl.
(4) Br_2 reacts with hot and strong NaOH solution to give NaBr, $NaBrO_3$ and H_2O.

CPB424

6. Which of the following oxyacid contains both P–H and P–P bond simultaneously?
(1) $H_4P_2O_5$ (2) $H_4P_2O_7$
(3) $H_4P_2O_6$ (4) None

CPB425

7. Which among the following order of given properties is incorrect
(1) $HOCl > HClO_2 > HClO_3 > HClO_4$
– oxidising nature
(2) $Cl > F > Br > I$
– electron affinity
(3) $Cl_2 > Br_2 > I_2 > F_2$
– bond dissociation energy
(4) $HF < HCl < HBr < HI$
– acidic nature

CPB426

8. Which of the following statement is incorrect regarding B_2H_6?
(1) On methylation it gives $B_2H(CH_3)_5$
(2) It has two $3C–2e^-$ bonds
(3) There can be maximum six atoms in a plane
(4) It has four $2C–2e^-$ bonds

CPB427

9. $Na_2B_4O_7 \cdot 10H_2O \xrightarrow{\text{Heat}} X + NaBO_2 + H_2O$, X is:
(1) Na_3BO_3
(2) $Na_2B_4O_7$
(3) B_2O_3
(4) B_4O_7

CPB428

10. Which of the following statements regarding orthoboric acid (H_3BO_3) is false?
(1) It acts as a weak monobasic acid
(2) It is soluble in hot water
(3) It has a planar structure
(4) It acts as a tribasic acid

CPB429

11. Which of the following is a lewis acid ?
(1) H_2CO_3 (2) CCl_4
(3) H_3BO_3 (4) BF_4^-

CPB430

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12. In SiF_6^{-2} and $SiCl_6^{-2}$ which one is known and why?

(1) SiF_6^{-2} because of small size of F

(2) SiF_6^{-2} because of large size of F

(3) $SiCl_6^{-2}$ because of small size of Cl

(4) $SiCl_6^{-2}$ because of large size of Cl

CPB431

13. What is formula for carbon suboxide?

(1) CO

(2) CO_2

(3) C_2O_4

(4) C_3O_2

CPB432

14. Find the incorrect match:

(1) Al_2Cl_6 : 3C-4e⁻ bond is present

(2) $Al_2(CH_3)_6$: All carbon atoms are sp^3-hybridized

(3) I_2Cl_6 : Nonplanar

(4) Al_2Br_6 : Nonpolar

CPB433

15. BCl_3 does not exist as dimer but BH_3 exists as dimer (B_2H_6) because:

(1) Chlorine is more electronegative than hydrogen

(2) There is $p_\pi - p_\pi$ back bonding in BCl_3, but BH_3 does not contain such multiple bonding

(3) Large sized chlorine atoms do not fit in between the small boron atoms, whereas small-sized hydrogen atoms fit between boron atoms

(4) None of these

CPB434

16. Which of the following structure is non-planar?

(1) XeO_4

(2) I_2Cl_6

(3) Sheet silicates

(4) Inorganic graphite layer

CPB435

17. Which of the following oxide is acidic in nature?

(1) B_2O_3 (2) Al_2O_3

(3) Ga_2O_3 (4) In_2O_3

CPB436

Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	4	4	4	4	3	1	3	4	3	1	4	3	3
Question	16	17													
Answer	3	1													

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Exercise - I
d and f-block Elements

1. Which of the following change is possible in Aqueous Medium-

(1) $2\,Cu^{+} \longrightarrow Cu^{2+} + Cu^{0}$

(2) $2Cu^{0} \xrightarrow[-3e^{-}]{} Cu^{+} + Cu^{+2}$

(3) $2Cu^{+2} \xrightarrow[+3e^{-}]{} Cu^{0} + Cu^{+}$

(4) $Cu^{+2} + e^{-} \longrightarrow Cu^{+}$

CDF145

2. Colourless ion is :-

(1) Cr^{+3} (2) Sc^{+3} (3) Cu^{+2} (4) CrO_4^{2-}

CDF146

3. Electronic configurations related to Praseodymium, Neodymium & Promethium are respectively-

(1) $4f^3 6s^2$, $4f^4 6s^2$ & $4f^6 6s^2$

(2) $4f^7 6s^2$, $4f^{13} 6s^2$ & $4f^6 6s^2$

(3) $4f^3 6s^2$, $4f^4 6s^2$ & $4f^5 6s^2$

(4) $4f^{14} 5d^1 6s^2$, $4f^{14} 6s^2$ & $4f^6 6s^2$

CDF147

4. In aqueous solution, Eu^{2+} ion acts as:

(1) An oxidizing agent

(2) A reducing agent

(3) Either (1) or (2)

(4) None

CDF148

5. Among the lanthanoids the one obtained by synthetic method is:

(1) Lu (2) Pm (3) Pr (4) Gd

CDF149

6. K_2MnO_4 can be converted into $KMnO_4$ by :

(1) Passing CO_2 gas

(2) By passing Cl_2

(3) Electrolytic oxidation

(4) All of these

CDF150

7. Which of the following does not contain transition metal?

(1) Haemoglobin (2) Vitamin B_{12}

(3) Cisplatin (4) Chlorophyll

CDF151

8. Which of the following does not give disproportionation reaction in acidic or basic medium?

(1) MnO_4^{-2} (2) HNO_2

(3) CrO_4^{-2} (4) Cl_2

CDF152

9. Which of the following is an interstitial carbide?

(1) SiC (2) WC (3) CaC_2 (4) Be_2C

CDF153

10. Lanthanoids for which +II and +III oxidation states are common is:

(1) La (2) Nd (3) Ce (4) Eu

CDF154

Exercise-I
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	**1**	**2**	**3**	**2**	**2**	**4**	**4**	**3**	**2**	**4**

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Exercise - II d and f-block Elements

1. Which of the following arrangements does not represent the correct order of the property stated against it?
(1) $V^{2+} < Cr^{2+} < Mn^{2+} < Fe^{2+}$: paramagnetic behaviour
(2) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$: ionic size
(3) $Co^{3+} < Fe^{3+} < Cr^{3+} < Sc^{3+}$: stability in aqueous solution
(4) Sc < Ti < Cr < Mn : number of oxidation states
CDF155

2. The yellow colour solution of Na_2CrO_4 changes to orange red on passing CO_2 gas due to the formation of :-
(1) CrO_5 (2) CrO_3
(3) $Na_2Cr_2O_7$ (4) Cr_2O_3
CDF156

3. What is incorrect about the reactions of $KMnO_4$ and oxalic acid
(1) CO_2 is formed
(2) decolourisation is fast in begining but become slow after some time
(3) Mn^{2+} is autocatalyst
(4) It is a redox change
CDF157

4. Which statement is correct
(1) Most common oxidation state of lanthanoid is +2
(2) HCl can be used to acidify $KMnO_4$ during redox reaction
(3) In presence of CO_2, orange dichromate solution changes to yellow chromate
(4) To separate Fe_2O_3 and Al_2O_3, NaOH can be used
CDF158

5. On addition of small amount of $KMnO_4$ to concentrated H_2SO_4, a green oily compound is obtained which is highly explosive in nature the compound is.
(1) Mn_2O_7 (2) MnO_2
(3) $MnSO_4$ (4) Mn_2O_3
CDF159

6. Interstitial compounds are formed when small atoms are trapped inside the crystal lattice of metals. Which of the following is not the characteristic property of interstitial compounds?
(1) They have high melting points in comparison to pure metals.
(2) They are very hard
(3) They retain metallic conductivity
(4) They are chemically very reactive
CDF160

7. Which is correct :-

Process	Result
(1) $(Ti(H_2O)_6)Cl_3$ is heated	Colour is intensified
(2) $CuSO_4$ reacts with KCN	$K_2[Cu(CN)_4]$ is formed
(3) $KMnO_4$ reacts with HF	MnF_6 is formed
(4) K_2MnO_4 reacts with CO_2	$KMnO_4$ (purple) and MnO_2 (brown) is formed

CDF161

8. Which of the following is incorrectly match?

Catalyst	Process
(1) V_2O_5	Contact process
(2) $CuCl_2$	deacon process
(3) Finely divided Fe	Vegetable oil to ghee
(4) $TiCl_4 + Al(CH_3)_3$	Zieglar Natta Catalyst

CDF162

9. Across the lanthanoid series, the basicity of the lanthanoid hydroxides:
(1) Increases
(2) Decreases
(3) First increases and then decreases
(4) First decreases and then increases
CDF163

Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	1	3	2	4	1	4	4	3	2

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Exercise - I

Coordination Compounds

1. The correct IUPAC name of $[Pt(NH_3)_2Cl_2]$ is
(1) Diamminedichloridoplatinum (II)
(2) Diamminedichloridoplatinum (IV)
(3) Diamminedichloridoplatinum (0)
(4) Dichloridodiammineplatinum (IV)

CCO327

2. When $CrCl_3.6H_2O$ is treated with excess of $AgNO_3$, 3 Mol of AgCl are obtained. The formula of the complex is:
(1) $[CrCl_3(H_2O)_3] \cdot 3H_2O$
(2) $[CrCl_2(H_2O)_4]Cl \cdot 2H_2O$
(3) $[CrCl(H_2O)_5]Cl_2 \cdot H_2O$
(4) $[Cr(H_2O)_6]Cl_3$

CCO328

3. Indicate the complex ion which shows geometrical isomerism.
(1) $[Cr(H_2O)_4Cl_2]^+$
(2) $[Pt(NH_3)_3Cl]$
(3) $[Co(NH_3)_6]^{+3}$
(4) $[Co(CN)_5(NC)]^{-3}$

CCO329

4. How many EDTA molecules are required to make an octahedral complex with a Ca^{+2} ion?
(1) 1
(2) 3
(3) 4
(4) 2

CCO330

5. In Cu-ammonia complex the state of hybridization of Cu^{+2} is
(1) sp^3
(2) spd^2
(3) sp^3d^2
(4) dsp^2

CCO331

6. Which complex has highest paramagnetism?
(1) $[Cr(H_2O)_6]^{+3}$
(2) $[Fe(H_2O)_6]^{+2}$
(3) $[Cu(H_2O)_6]^{+2}$
(4) $[Zn(H_2O)_6]^{+2}$

CCO332

7. Which of the following statement is correct about ethane 1,2 diamine?
(1) It is a neutral ligand
(2) It is a bidentate ligand
(3) It is a chelating ligand
(4) All of the above

CCO333

8. Which of the following have maximum number of isomers
(1) $[Co(NH_3)_4Cl_2]^+$
(2) $[Ni(en)(NH_3)_4]^{+2}$
(3) $[Ni(C_2O_4)(en)_2]^0$
(4) $[Cr(SCN)_2(NH_3)_4]^+$

CCO334

9. Amongst the following the most stable compound is
(1) $[Fe(H_2O)_6]^{3+}$
(2) $[Fe(C_2O_4)_3]^{3-}$
(3) $[Fe(NH_3)_6]^{3+}$
(4) $[FeCl_6]^{3-}$

CCO335

10. Hexafluoroferrate(III) ion is an outer orbital complex, the number of unpaired electrons present in it is
(1) 1
(2) 5
(3) 4
(4) unpredictable

CCO336

11. Which of the following species can act as reducing agent ?
(1) $[Co(CO)_4]^-$
(2) $[Mn(CO)_6]$
(3) $[Mn(CO)_5]$
(4) $[Cr(CO)_6]$

CCO337

12. $[NiCl_4]^{-2}$ and $[Ni(CN)_4]^{-2}$ resemble in
(1) geometry
(2) magnetic nature
(3) hybridisation
(4) oxidation state

CCO338

13. CFSE of high spin d^4 complex is
(1) $-0.6\ \Delta_0$
(2) $-1.8\ \Delta_0$
(3) $-1.6\ \Delta_0 + P$
(4) $-1.2\ \Delta_0$

CCO339

14. A brown ring is formed in the ring test of NO_3^- ion. It is due to the formation of
(1) $[Fe(H_2O)_5(NO)]^{2+}$
(2) $FeSO_4.NO_2$
(3) $[Fe(H_2O)_4(NO)_2]^{2+}$
(4) $FeSO_4.HNO_3$

CCO340

15. Hybridization, shape, and magnetic moment of $K_3[Co(CO_3)_3]$ are:
(1) d^2sp^3, octahedral, 4.9 B.M.
(2) sp^3d^2, octahedral, 4.9 B.M.
(3) dsp^2, square planar, 4.9 B.M.
(4) sp^3, tetrahedral, 4.9 B.M.

CCO341

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Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	1	1	4	2	4	4	2	2	2	4	1	1	2

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Exercise - II Coordination Compounds

1. Geometrical shapes of the complexes formed by the reaction of Ni^{+2} with Cl^-, CN^- and H_2O respectively are.
(1) Octahedral, tetrahedral and sq. planar
(2) Tetrahedral, square planar and octahedral
(3) Square planar, tetrahedral and octahedral
(4) Octahedral, tetrahedral and square pyramidal

CCO342

2. Which of the following doesn't follow effective atomic number rule?
(1) $[Cu(NH_3)_4]^{+2}$
(2) $[Zn(OH)_4]^{-2}$
(3) $[HgI_4]^{2-}$
(4) $Fe(CO)_5$

CCO343

3. Same number of unpaired electron is observed in which of the following complexes
(a) $[MnCl_6]^{3-}$ (b) $[Fe(CN)_6]^{3-}$
(c) $[CoF_6]^{3-}$ (d) $[Ni(NH_3)_6]^{2+}$
(1) a and b (2) a & c
(3) b & d (4) c & d

CCO344

4. When excess of ammonia is added to $CuSO_4$ solution. The deep blue colour complex is formed. The complex is
(1) Tetrahedral & paramagnetic
(2) Tetrahedral & diamagnetic
(3) Square planar & diamagnetic
(4) Square planar & paramagnetic

CCO345

5. Select the correct I.U.P.A.C. name for $[Cr(C_6H_6)(CO)_3]$.
(1) (η^6-benzene) tricarbonylchromate (0)
(2) Tricarbonyl (η^6-benzene) chromate (0)
(3) Tricarbonyl (η^6-benzene) chrominum (0)
(4) (η^6-benzene) tricarbonylchromium (0)

CCO346

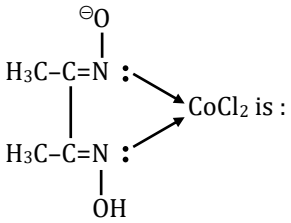
6. Which of the following statement is correct?
(1) With d^2sp^3 hybridization $[FeCl(CN)_4(O_2)]^{4-}$ complex is diamagnetic
(2) $[NiCl_4]^{2-}$ complex is more stable than $[Ni(dmg)_2]$
(3) $[V(CO)_6]$ is not very stable and easily reduces to $[V(CO)_6]^-$
(4) Ligands such as CO, CN^-, NO^+ are pie– donar due to the presence of filled p-molecular orbital

CCO347

7. Mg is an important component of which biomolecule occurring extensively in living world?
(1) Hemoglobin (2) Chlorophyll
(3) Vit B_{12} (4) ATP

CCO348

8. The correct IUPAC name of the complex:



(1) dichloridodimethylglyoximatocobalt (III)
(2) bis(dimethylglyoxime) dichlorido cobalt (II)
(3) dimethylglyoximecobalt (II) chloride
(4) dichloridodimethylglyoxime-N,N-cobalt (II)

CCO349

9. Among $[Ni(CN)_4]^{2-}$,$[NiCl_4]^{2-}$ and $[Ni(CO)_4]$:
(1) $[NiCN)_4]^{2-}$ is square planar and $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ are tetrahedral
(2) $[NiCl_4]^{2-}$ is square planar and $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$ are tetrahedral
(3) $[Ni(CO)_4]$ is square planar and $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ are tetrahedral
(4) None of these

CCO350

10. Which ion is paramagnetic?
(1) $[Ni(H_2O)_6]^{2+}$
(2) $[Fe(CN)_6]^{4-}$
(3) $[Ni(CO)_4]$
(4) $[Ni(CN)_4]^{2-}$

CCO351

11. Which is used in cancer-chemotherapy ?
(1) Cis-platin
(2) Zeise's salt
(3) both 1 and 2
(4) None of these

CCO352

12. Spin only magnetic moments of a d^8 ion in an octahedral, square planar and tetrahedral complex, respectively are
(1) 2.8 BM, 0.4 B.M., 2.8 BM
(2) 2.8 BM, 0 BM, 2.8 BM
(3) 0, 0 & 0 B.M.
(4) None of these

CCO353

13. Select the correct matching in given table –

	Coordination number	Type of hybridisation	Hybrid orbital in space
(1)	4	dsp^2	square planar
(2)	4	sp^3	square planar
(3)	6	sp^3d	octahedral
(4)	5	sp^3d	octahedral

CCO354

14. The IUPAC name for ionisation isomers of $[Pt(NH_3)_3(Br)(NO_2)(I)]$ Cl is
(1) Triamminebromidochloridonitro platinum (IV) iodide
(2) Triamminebromidochloridoiodido platinum(IV) nitrite
(3) Triamminechloridoiodidonitro platinum (IV) bromide
(4) All are possible

CCO355

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Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Answer	2	1	2	4	4	3	2	1	1	1	1	2	1	4